minkowski-correlations

API Reference | Version 0.1.0

minkowski_functionals()

```
minkowski_functionals(
    data_points: numpy.ndarray,
    inner_radius: float,
    outer_radius: float,
    window: tuple[float, float, float, float] | None = None,
    boundary_condition: str = 'none',
    include_boundary: bool = False,
    jobs: int = -1,
    return_geometry: bool = False,
)
```

Description

Compute the Minkowski functionals of a union of annuli.

Each input point defines an annulus with a common inner and outer radius. The Minkowski functionals are evaluated analytically from the exposed boundary arcs, including contributions from the observation window where applicable.

Parameters

data_points : numpy.ndarray

Point coordinates as an array of shape (N, 2). Duplicate points are automatically removed while preserving the order of their first occurrence.

inner_radius : float

Inner radius of the annuli. Must be nonnegative and smaller than outer_radius. A value of zero produces disks.

outer_radius : float

Outer radius of the annuli. Must be positive.

window : tuple of float or None, optional

Rectangular observation window given by (x_min, x_max, y_min, y_max). Required for "periodic" and "minus" boundary conditions. Ignored when boundary_condition is "none". Default is None.

boundary_condition : {"none", "periodic", "minus"}, optional

Method used to handle edge effects caused by the finite observation window: - "none": Ignore the observation window and compute the Minkowski functionals of the full union of annuli. - "periodic": Apply periodic boundary conditions, treating the rectangular observation window as a periodic unit cell. Annuli crossing one boundary re-enter from the opposite side. - "minus": Apply minus sampling by shrinking each side of the rectangular observation window by outer_radius. Points outside the reduced window are retained to account for their contributions inside it. The default is "none". Periodic boundary conditions and minus sampling require an axis-aligned rectangular window.

include_boundary : bool, optional

If True and boundary_condition is "minus", include the contributions of the covered observation-window boundary to the perimeter and Euler characteristic. By default, these contributions are excluded.

jobs : int, optional

Number of parallel workers used by joblib. A value of -1 requests all available workers. Default is -1.

return_geometry : bool, optional

If True, return the geometric intermediate results in addition to the Minkowski functionals. Default is False.

Returns

MF : list of float

Minkowski functionals in the order (area, half the perimeter, Euler characteristic), corresponding to (M0, M1, M2).

innerCircles_new : list, optional

Inner circle boundaries after intersection with the observation window. Returned if return_geometry is True.

outerCircles_new : list, optional

Outer circle boundaries after intersection with the observation window. Returned if return_geometry is True.

Rectangle : RectangleSurface, optional

Accumulated observation-window boundary segments. Returned if return_geometry is True.

points : list, optional

Point coordinates used in the calculation, including periodic copies where applicable. Returned if return_geometry is True.

Raises

ValueError

If the point coordinates, radii, observation window, or boundary condition are invalid.

TypeError

If the input points cannot be converted to numeric values.

Notes

The implementation evaluates boundary integrals over the exposed arcs of the union of annuli. The Euler characteristic is obtained from curvature and vertex contributions.